Kakeya Records: an AI agent sets new best-known thin-triangle Kakeya constructions in one hour on a laptop

Shrestha Pandey · 4th Open Scientific Intelligence Hackathon · solo entry, directing Claude Code (Claude Opus 5.5) on an 8 GB Apple M1 · code and certificates: github.com/shrestha-droid/kakeya-records

Problem. Take N = 128 directions $a_i = i/N$. For each one, place a thin triangle with vertical base $[b_i-\delta, b_i]$ at x = 0 ($\delta = 1/N$) and apex $(1, a_i+b_i)$. Choose the intercepts b_i to *minimise the area of the union*. This discretised Kakeya needle problem is open: Keich (1999) gave the asymptotically optimal 1/log N construction, but the best finite configurations are unknown. It is HorizonMath problem `keich_thin_triangles_128`. That benchmark collects open problems whose candidate solutions can be checked automatically, and a strong frontier model (GPT-5.4 Pro) set the previous benchmark result there.

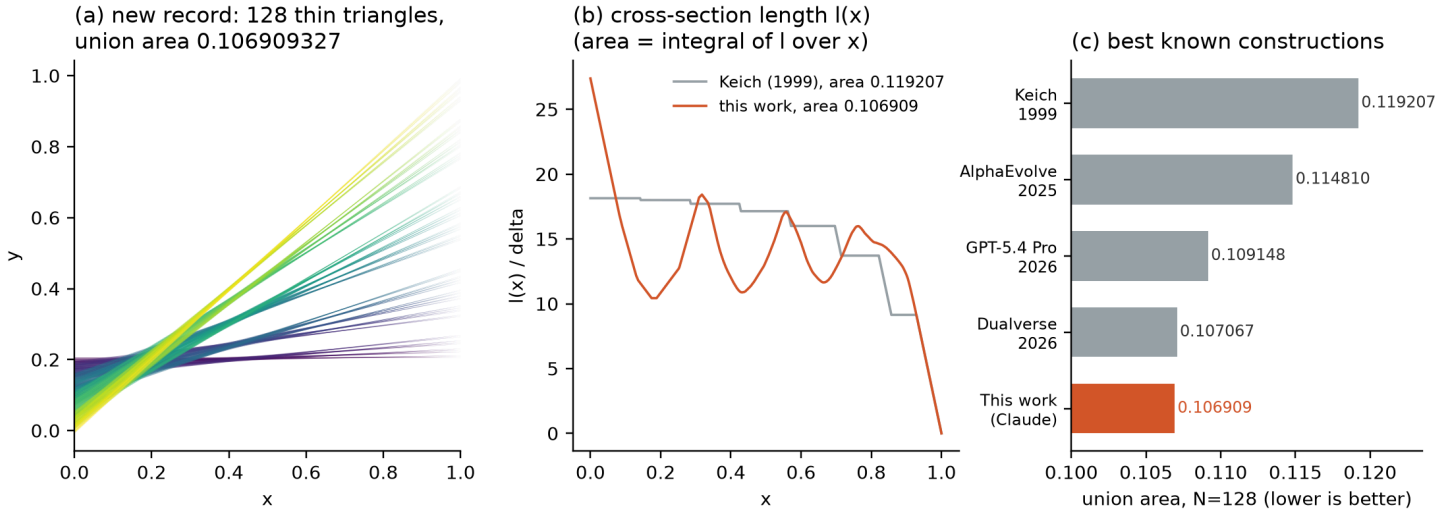


Figure. (a) The new N = 128 construction. (b) Slice length l(x), whose integral is the area. Keich's profile is a flat staircase; ours pays more at x = 0 so that triangle bundles can converge at three focal heights (x ≈ 0.17, 0.44, 0.66). (c) Best constructions over time.

Results (union area, lower is better)

N	AlphaEvolve 2025	Prev. best 2026	This work
32	0.141647	0.138596	**0.138432**
64	0.121735	0.120901	**0.120839**
128	0.114810	0.107067	**0.106909327**
256	–	– (Keich 0.106832)	**0.096005**

At N = 128 the new value is 0.15% below the previous best (Dualverse Station), 2.05% below GPT-5.4 Pro (HorizonMath), 6.88% below AlphaEvolve and 10.32% below Keich. Two independent checks certify every value. One is the benchmark's official validator. The other is our verifier, which computes the *exact rational* area of the stored configuration: it converts every float exactly, then integrates the piecewise-linear slice length exactly over all 32,640 breakpoints.

Agentic workflow

I led the project and directed the agent toward a verifiable record within the event. The agent surveyed the 113 HorizonMath problems, read the validators, and picked a continuous construction problem where a laptop could compete. It then derived a fast formulation, checked the literature (finding the stronger 2026 value), and iterated. It beat AlphaEvolve and GPT-5.4 Pro 7 minutes after starting, and set the record at minute 63.

Method

- **Exact slice formula.** All N intervals in a slice have the same length $L = \delta(1-x)$, so $l(x) = L + \Sigma \min(g_k, L)$ over gaps between sorted tops. That gives the area and an exact subgradient with no geometry library.
- **Annealed soft-min + Adam.** A smoothed min with temperature going to 0 lets gradients escape the piecewise-linear kinks.
- **Lifting lemma.** A triangle at resolution N is exactly the union of two resolution-2N children (intercepts b and b−1/2N). Optimal area is therefore non-increasing under dyadic refinement, which justifies a coarse-to-fine beam search from N = 8 to 128.
- **Memetic islands.** Mutations (noise, dyadic crossover, block shift, tilt, reset) are each followed by descent, with migration between islands. A Numba kernel uses insertion sort warm-started across neighbouring slices for about 4× throughput.
- **Symmetry.** The problem is invariant under $b_i \to \delta - b_{N-1-i}$, yet optima are asymmetric.

Significance and limits

A single agent with a few CPU-hours beat a frontier-model ensemble (AlphaEvolve) by 6.8%, and the best previous agentic result by a smaller margin. Reproducing it takes one minute with the provided verifier. These are upper bounds, and optimality is open. The previous best comes from a 2026 web notebook, not a journal. Next steps: lower bounds via LP duality, and asymptotic families extrapolated from the focal structure.